

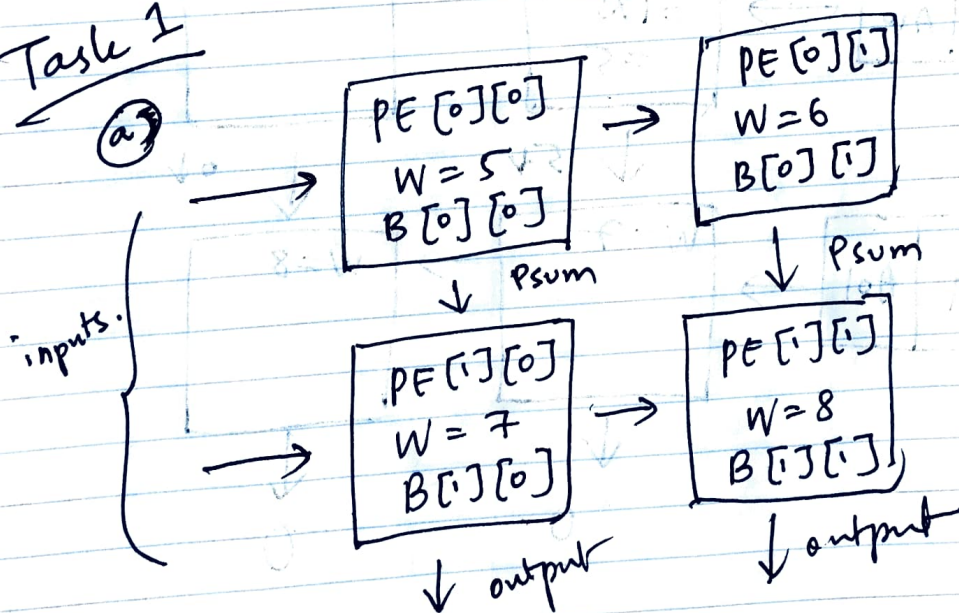
Codefest-5

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

$$B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$$

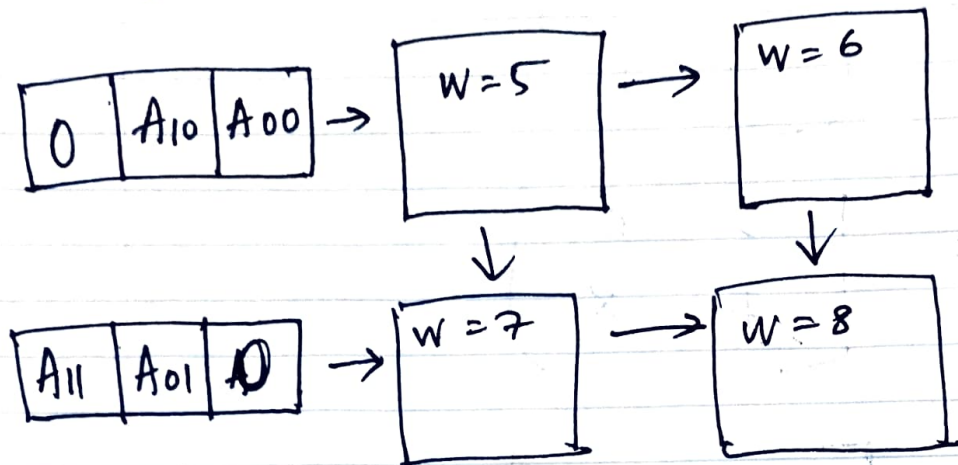
$$C = \begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix}$$

Task 1

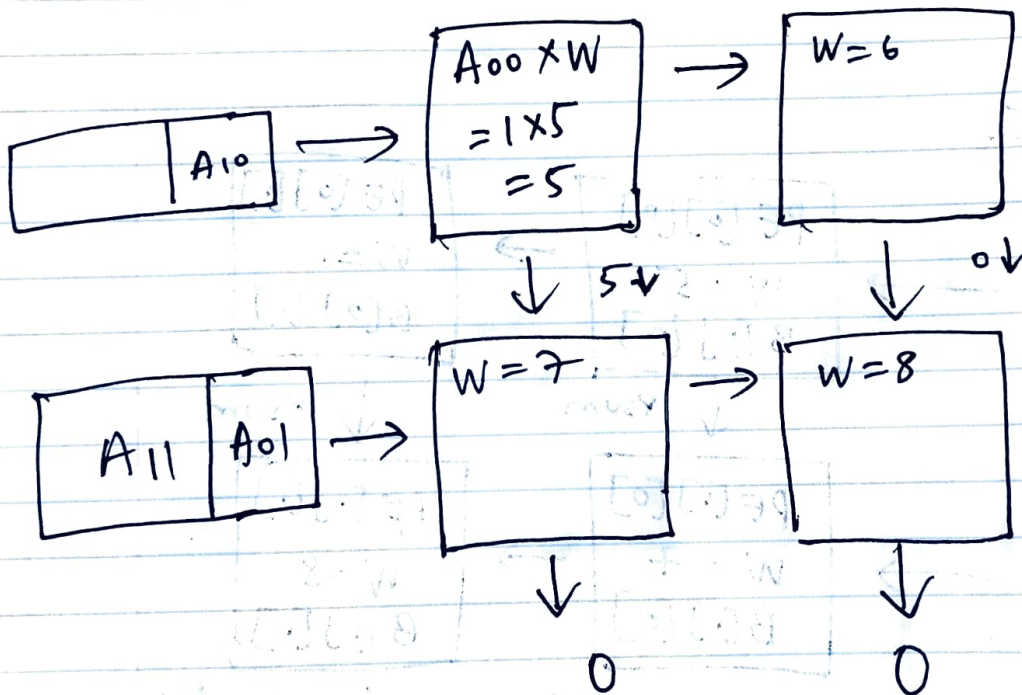


Task-2

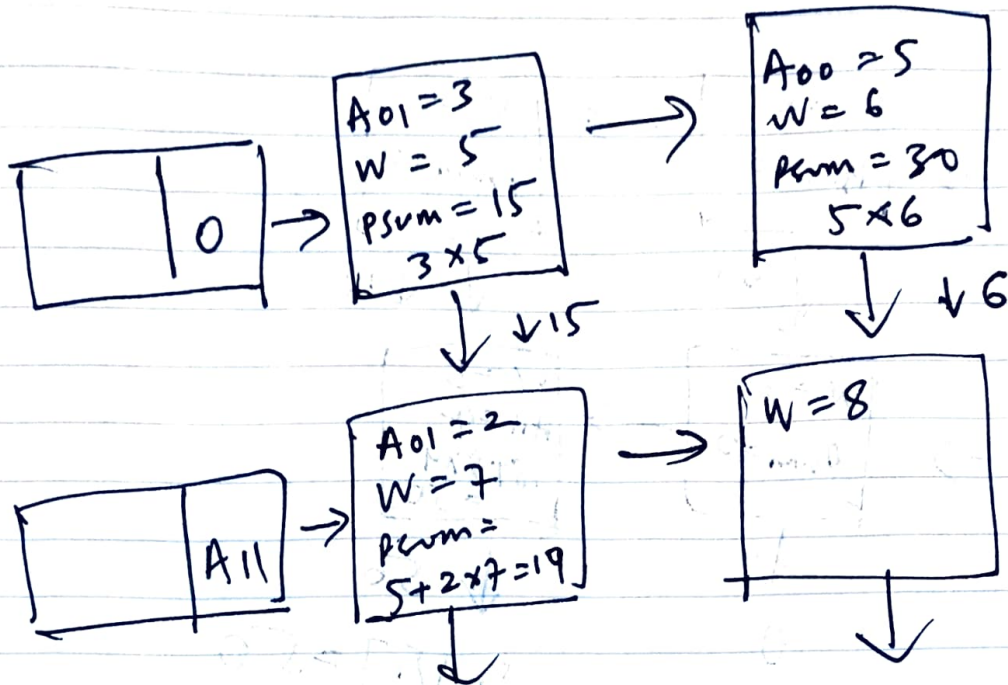
$t = 0$



$t = 1$



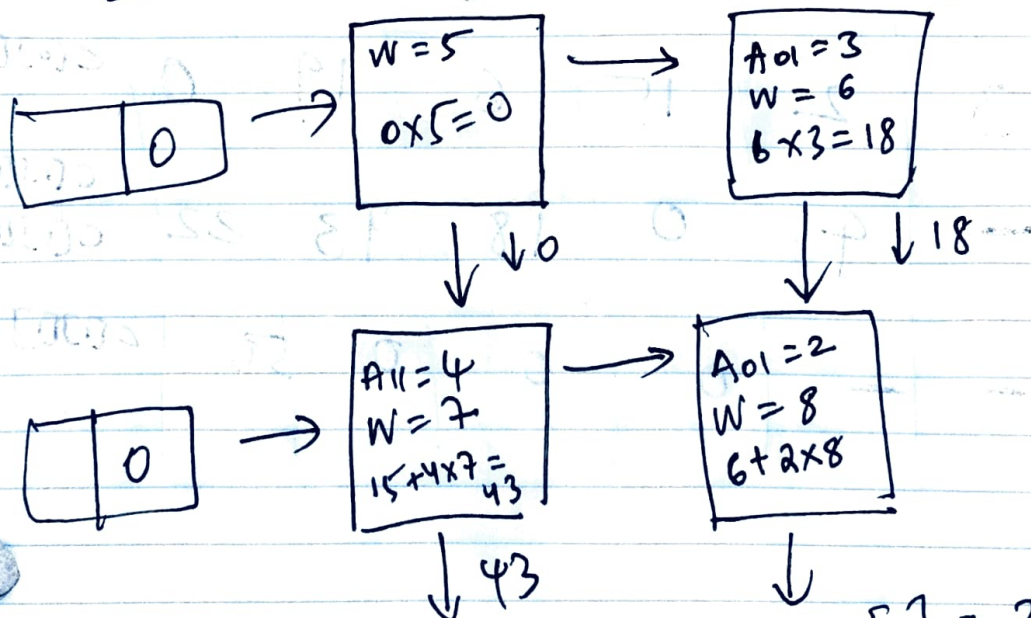
t=2



$$c[0][0]=19$$

0

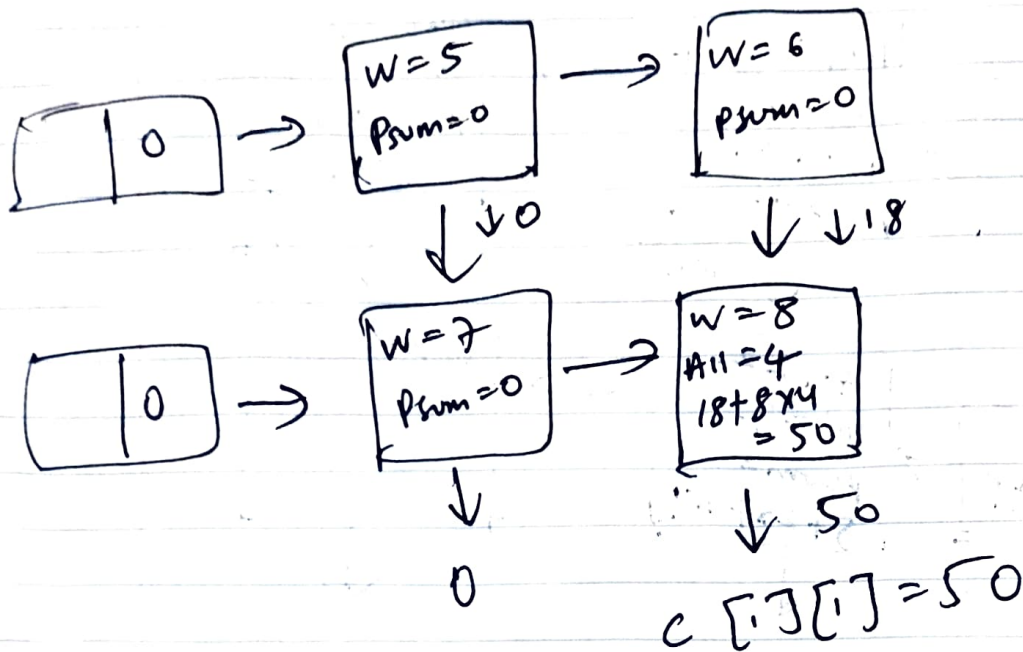
t=3



$$c[1][0]=43$$

$$c[0][1]=22$$

t=4



Summary

cycle	Row 0	Row 1	$PE[0][0]$	$PE[0][1]$	$PE[1][0]$	$PE[1][1]$	O/p.
1	1	—	5	0	0	0	—
2	3	2	15	6	19	0	$C[0][0]=19$
3	—	4	0	18	43	22	$C[1][0]=43$ $C[0][1]=22$
4	—	—	0	0	0	50	$C[1][1]=50$

Task 3

MAC count

Input reuse count

OFF-chip access count for A, B, C

(a) Total MAC count

$$4 \text{ PEs} \times 2 \text{ inputs each} \\ = 8 \text{ MACs.}$$

(b) Input reuse

For a $N \times N$ weight stationary matrix an input is fetched once and reused $N-1$ times.

For 2×2 array an input is reused $2-1 = 1$ time.

So they are fetched once, used twice.

(c) ~~Accessing~~ the fetched data of the inputs get streamed while the input weights stay stationary.

Matrix	Access Type	Accesses.
A	Read	4
B	Read	4
C	Write	4

Total 12 off chip accesses.

Task 4 Output-Stationary:

Partial sums (outputs) stay
fixed in the PEs.